Total HW: 2.18, 22, 29, 30

**Problem 2.18.** Use Stirling's approximation to show that the multiplicity of an Einstein solid, for any large values of $N$ and $q$, is approximately

$$\Omega(N,q) \approx \frac{\left(\frac{q+N}{q}\right)^q \left(\frac{q+N}{N}\right)^N}{\sqrt{2\pi q(q+N)/N}}.$$

The square root in the denominator is merely large, and can often be neglected. However, it is needed in Problem 2.22. (Hint: First show that $\Omega = \frac{N}{q+N}\frac{(q+N)!}{q!\,N!}$. Do not neglect the $\sqrt{2\pi N}$ in Stirling's approximation.)

$$\Omega(N,q) = \frac{(q+N-1)!}{q!\,(N-1)!}$$

$$\Omega(N,q) = \frac{N}{q+N}\cdot\frac{(q+N)!}{q!\,N!}$$

Now, apply Stirling's approx

$$N! = N\cdot\underbrace{(N-1)\cdot(N-2)\cdot\ldots\cdot 1}_{(N-1)!}$$

$$N! = N\cdot(N-1)!$$

$$(N-1)! = \frac{N!}{N}$$

$$(q+N-1)! = \frac{(q+N)!}{(q+N)}$$

$$N! = \left(\frac{N}{e}\right)^N\cdot\sqrt{2\pi N} = N^N e^{-N}\cdot\sqrt{2\pi N}$$

test Stirling's w/ python

$$\Omega = \frac{N}{q+N} \frac{\left(\frac{q+N}{e}\right)^{q+N} \sqrt{2\pi (q+N)}}{\left(\frac{q}{e}\right)^{q} \sqrt{2\pi q} \left(\frac{N}{e}\right)^{N} \sqrt{2\pi N}}$$

$$= \frac{(N)^{2/2}}{(q+N)^{2/2}} \frac{(q+N)^{q} (q+N)^{N} (q+N)^{1/2}}{q^{q} \cdot N^{N} (2\pi q N)^{1/2}}$$

$$= \frac{\left(\frac{q+N}{q}\right)^{q} \left(\frac{q+N}{N}\right)^{N}}{\sqrt{2\pi q (q+N)/N}}$$

2.22 (a) $q = 2N$

$N_A = N_B = N$

macrostates $= q + 1 = 2N + 1$

$0 \to 2N$

(b) total number of microstates → single Einstein solid w/ total number of particles and energy → $N \to 2N$, $q \to 2N$

$$\Omega = \frac{\left(\frac{q+N}{q}\right)^q \left(\frac{q+N}{N}\right)^N}{(2\pi q(q+N)/N)^{1/2}}$$

$$= \frac{\left(\frac{4N}{2N}\right)^{2N}\left(\frac{4N}{2N}\right)^{2N}}{(2\pi 2N(4N)/2N)^{1/2}}$$

$$\Omega = \frac{2^{4N}}{\sqrt{8\pi N}} \quad \checkmark$$

(c) $\Omega_{max} = \Omega_A(q=N, N=N)\,\Omega_B(q=N, N)$

$$\Omega_{max} = \left(\frac{\left(\frac{2N}{N}\right)^N\left(\frac{2N}{N}\right)^N}{(2\pi N(2N)/N)^{1/2}}\right)^2$$

$$= \left(\frac{2^{2N}}{(4\pi N)^{1/2}}\right)^2 = \frac{2^{4N}}{4\pi N} \quad \checkmark\checkmark$$

(d) part (c) gives height

part (b) gives area

$$\text{width} = \frac{\text{part (b)}}{\text{part (c)}} = \frac{4\pi N}{\sqrt{8\pi N}} = \frac{4\pi}{\sqrt{8\pi}} \cdot N^{1/2}$$

$$= \frac{4\pi\sqrt{8\pi}}{8\pi^2}\sqrt{N} = \frac{2\sqrt{2\pi N}}{2}$$

$$\text{width} = \sqrt{2\pi N}$$

total macrostates → $2N+1$

$\frac{\sqrt{2\pi N}}{2N+1}$ ← fraction w/ large probabilities

$N = 10^{23} \to \frac{\sqrt{2\pi 10^{23}}}{2\cdot 10^{23}} = \frac{\sqrt{2\pi}}{2} \cdot 10^{-11.5}$

$= 0.79 \cdot 10^{-11.5}$

$= 2.5 \cdot 10^{-12}$

fraction of macrostates w/ reasonable probability to occur.

2.29 $\Omega_{total} = 6.9 \cdot 10^{114}$ (most likely) $\quad \Omega_{total} = 2.8 \cdot 10^{91}$ (least likely) $\quad \Omega_{total} = 9.3 \cdot 10^{115}$ (all microstates)

$\frac{S}{k_B} = 264 \quad \frac{S}{k_B} = 187.5 \quad \frac{S}{k_B} = 267$

2.30 (a) all microstates $\rightarrow \frac{2^{4N}}{\sqrt{8\pi N}} \qquad N = 10^{23}$

$$\frac{S}{k_B} = \ln\left(2^{4\cdot 10^{23}}\right) - \frac{1}{2}\ln\left(8\pi \cdot 10^{23}\right)$$

$$= \underbrace{4\cdot 10^{23}\ln(2)}_{2.8\cdot 10^{23}} - \underbrace{\frac{1}{2}\ln\left(8\pi\cdot 10^{23}\right)}_{28}$$

$$\frac{S}{k_B} = 2.8\cdot 10^{23} - 28$$

(b) most likely macrostate $\rightarrow \frac{2^{4N}}{(4\pi N)}$

$$\frac{S}{k_B} = 4\cdot 10^{23}\ln 2 - \underbrace{\ln\left(4\pi\, 10^{23}\right)}_{55}$$

$$\frac{S}{k_B} = 2.8\cdot 10^{23} - 55$$ ← smaller than part (a)! but not by much